Why Bogs Preserve Bodies for Thousands of Years

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In August 1984, **peat** cutters near Wilmslow, south of Manchester, England, found a body in a **peat** bog. Known as the Lindow Man, it had lain there for nearly 2,000 years, yet was still recognizably human. He is one of at least 2,000 bog bodies recovered from Europe's peatlands, dating from the **Mesolithic** period, which ended in Europe between 8,500 and 5,500 years ago, through to World War II.

A **peat** bog is an acidic, nutrient-poor wetland made of **peat**, soft, spongy ground consisting of partly decayed plant material building up over hundreds or thousands of years. **Peat** is organic matter that builds up faster than it decays, but it mostly consists of **sphagnum** moss, a moisture-loving moss that stores a lot of water.

That waterlogged moss leaves little air in the bog, the first key to preservation. Bogs get almost all their water from rain and snow, so they are low in minerals and nutrients, and the water barely moves. Thick layers of decomposed material choke off the air below. Without oxygen, the environment becomes highly acidic, and bacteria and fungi that would normally break down dead material cannot thrive.

Instead, the high acidity essentially pickles a body, like a brine stopping harmful bacteria from growing. Author Anna North describes preserved bodies as living time capsules. **Sphagnum** moss also releases a carbohydrate called **sphagnan**, which binds to proteins in skin and tissue, turning skin leathery and dark-brown, like tea, while protecting it against decay. Skin, hair, nails, and internal organs can therefore remain remarkably intact for thousands of years.

Bones fare differently: acid **leaches** calcium from them, so they can soften, warp, or dissolve completely. Raised bogs, deep dome-shaped peatlands wetter and higher than surrounding land, preserve bodies best because they are highly acidic and entirely without oxygen.

Word Treasure

peat -- Peat is soft, spongy ground made of partly decayed plant material.

sphagnum -- Sphagnum moss is a moisture-loving moss that stores a lot of water.

sphagnan -- Sphagnan is a carbohydrate from sphagnum moss that binds to proteins in skin and tissue.

Mesolithic -- The Mesolithic period is a very early period in human history.

leaches -- To leach something is to wash a substance out of it.

Background you need

A peat bog is a wetland where dead plant material builds up faster than it breaks down, forming soft, spongy ground.

Because the ground is waterlogged and still, little air reaches below the surface, and the water becomes highly acidic.

Without oxygen, the bacteria and fungi that normally break down dead material cannot thrive, so soft tissue can survive far longer than usual.

That is why bogs can preserve skin, hair, and organs, even though their acid weakens bones by removing calcium.

Break it down

LEAD: A body found in a peat bog in 1984 had lain there for nearly 2,000 years yet was still recognizably human.

KEY EVENT: Scientists explain how peat bogs preserve bodies for thousands of years.

+- WHAT: A peat bog is an acidic wetland made of peat, mostly sphagnum moss

+- HOW: Waterlogged moss leaves little air, so the bog becomes highly acidic and bacteria cannot thrive

+- RESULT: Acidity pickles soft tissue, and sphagnan protects skin, while acid weakens bones

EVIDENCE: Skin, hair, nails, and internal organs can remain intact, but bones can soften or dissolve

+- BEST CASE: Raised bogs preserve bodies best because they are highly acidic and entirely without oxygen

Why it matters

Because bogs can preserve bodies for thousands of years, they give scientists a rare window into people and times that would otherwise be lost.

Test what you remember

1. What is a peat bog?

- (A) A mountain lake
- (B) An acidic, nutrient-poor wetland made of peat
- (C) A dry, sandy desert
- (D) A deep ocean trench

2. Who was found in a peat bog in August 1984 near Wilmslow?

- (A) A peat cutter
- (B) Anna North
- (C) A sphagnum plant
- (D) The Lindow Man

3. Why cannot bacteria and fungi thrive in a bog?

- (A) Because the bog is highly acidic and without oxygen
- (B) Because bogs are too hot
- (C) Because bogs have too much air
- (D) Because peat is too dry

4. What does sphagnan do?

- (A) It adds oxygen to the bog
- (B) It turns peat into sand
- (C) It dissolves bones
- (D) It binds to proteins in skin and tissue, protecting them against decay

5. What happens to bones in a bog?

- (A) They turn to gold
- (B) They grow stronger
- (C) They stay exactly the same
- (D) Acid leaches calcium, so they can soften, warp, or dissolve

6. Which bogs preserve bodies best?

- (A) Bogs low in acid
- (B) Bogs with moving water
- (C) Raised bogs, which are highly acidic and entirely without oxygen
- (D) Bogs full of air

Answer key (try the quiz first!): 1: B 2: D 3: A 4: D 5: D 6: C

Questions to think about

1. Anna North: She describes preserved bodies as living time capsules.

MY THOUGHTS (at least 20 words):